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A Generalized Voronoi Diagram based Robot Exploration Method for Mobile Robots

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***Abstract*—The Rapidly-exploring Random Tree (RRT) based method has been widely used in the field of autonomous exploration of mobile robots. During autonomous exploration, the robot relies on the RRT tree growth to obtain exploration frontiers. However, due to the trap space problem, such as narrow corridors and mazes, the RRT tree cannot grow to these regions in a limited time, resulting in the slow extraction of frontiers. Moreover, limited to the randomness of the growth of the RRT tree, the frontiers extracted from the RRT treetop have a lot of redundancy. In this paper, we propose a Generalized Voronoi Diagram (GVD) based exploration method to guide robots for efficient exploration. First, a lightweight feature extraction is proposed to extract the GVD information and represent the topological structure of the environment. Second, the GVD features are utilized to quickly generate heuristic frontiers. In order to reduce the redundancy of frontiers, a GVD frontiers fusion and extraction algorithm is proposed. Finally, the GVD nodes are fused and we obtain a feature node set. By using the collision-check module, we get the feature matrix. The path cost is quickly calculated by using the feature matrix. The experimental results show that by comparing with the RRT-Exploration algorithm, our method has better performance in quickly extracting exploration frontiers and reducing backtracking.**

***Index Terms*—Robot Exploration, GVD, Frontier Detection, GVD Feature Matrix, Path Cost.**

I. Introduction

Efficient autonomous exploration in a limited known environment is a challenging issue for mobile robots. Autonomous exploration allows robots to automatically obtain an accurate map of an unknown environment and navigate to a goal [1]. Recently, frontier-based robotic exploration methods [2]–[4] have been widely used, in which frontiers are utilized to divide the exploration space into known and unknown areas, and guide the robot to collect the information in the unknown area of the map. Due to the lack of prior information about the environment, the robot needs to constantly detect the frontiers. Many academics in the field of autonomous exploration prefer the Rapidly-exploring Random Tree (RRT) [5], which can bypass the complex creation of the configuration space by integrating a collision-check module in the sampling-based algorithm. RRT based exploration method records the place of the tree top as the frontier. However, as shown

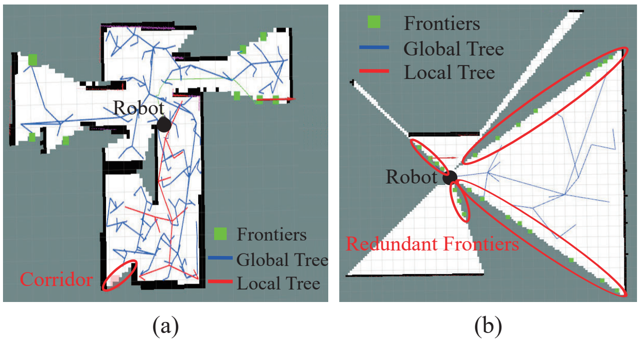

Fig. 1. (a) the RRT tree cannot grow to the boundary to obtain frontiers in red oval area. (b) the redundancy of the frontiers extracted by the RRT-Exploration at the boundary in red oval area. The black circle is the robot current position in the environment, and the green squares represent the candidate exploration frontiers. The local tree is represented by the red lines, the global tree is represented by the blue lines.

in the red oval area of Fig. 1(a), robot may be stuck in several challenging environments, such as mazes, narrow passages and corridors, *etc.*, and it is difficult for RRT trees to grow to the boundary within a limited time period, resulting in the backtracking phenomenon and even the failure of autonomous exploration. Furthermore, due to the randomness of the growth of the RRT tree, the frontiers extracted by the RRT based exploration method have a lot of redundancy, as shown in the red oval area of Fig. 1(b), resulting in time-consuming in the robot decision-making process.

In order to address the aforementioned problems, we propose to utilize the Generalized Voronoi Diagram (GVD) [6] to generate heuristic frontiers and guide the autonomous exploration of robots. Herein, we draw inspiration from GVD, which consists of the set of nodes which are equally spaced apart from two or more obstacles. GVD can represent the topological structure of the environment with only a few nodes, and this representation method is relatively concise and complete even in the trap area. Therefore, in this work, we propose to utilize GVD instead of RRT for frontier extraction. To further the reduce redundancy of the frontiers, we propose a frontier fusion algorithm to remove unnecessary frontiers, which greatly reduces the computational complexity of frontier selection and guarantees that the exploration is performing in real time. During the robot decision-making process, we propose a GVD feature nodes fusion algorithm to calculate the path cost, reducing the backtracking phenomenon in the robot exploration process.

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The remainder of this paper is structured as follows. We first discuss related research on frontier-based robotic exploration methods in Section II. In Section III, we go to great lengths to describe the GVD-based exploration strategy. We carry out a number of experiments and analyze the results in Section IV, and then reach conclusions in Section V.

II. RELATED WORK

Frontier-based exploration is one of the most common robot autonomous exploration approaches and the key problem is how to extract and select frontiers efficiently [7]–[10]. Recently, numerous scholars have proposed various strategies to expedite the extraction of frontiers. To extract frontiers, Umari *et al.* [11] proposed using RRT [5], and the nodes where the RRT tree grows to the boundary are marked as prospective frontiers. Although sampling-based methods avoid the construction of the complex spaces, due to the trap spaces, RRT trees cannot grow to these regions within a limited time in some complex environments, resulting in the low efficiency of autonomous exploration. Keida and Kaminka [8] combined Wavefront Frontier Detector (WFD) and Fast Frontier Detector (FFD) to speed up frontiers extraction, in which WFD checks known areas and FFD checks new laser reading data. Sun *et al.* [12] proposed an integrated frontiers detection and a maintenance method that gradually discovers potential frontiers by locally updating the distance map. Although these methods can improve the performance of frontiers extraction to a certain extent, they consume a large amount of computing resources.

Another challenge faced by the frontier-based methods is how to select the frontier as the exploration goal according to the limited known environmental knowledge and the current position of the robot. Yamauchi *et al.* [2] chose the frontier closest to the robot as the exploration target. Gonzalez-Banos and Latombe [13] proposed a utility function to evaluate the frontiers, which integrated the information gain of the frontier itself and the distance between the frontiers and the position of the robot. Stachniss *et al.* [14] presented a decision-theoretic framework to evaluate the cost by considering the uncertainty and expected information gain. Umari *et al.* [11] improved the information gain in the utility function, taking into account unknown grids within the bounds of a predetermined radius, and incorporate path costs. Liu *et al.* [15] proposed heuristic exploration by using prior knowledge of the environment. However, most of the frontiers are redundant, and it takes a lot of time to evaluate these redundant points. In addition, when evaluating the motion cost of the robot from current pose of the robot to the target frontier, two methods are usually used: one is to estimate the straight-line distance between two points, and the other is to calculate the specific path length by using the path planning method. Umari *et al.* [11] use Euclidean Distance to evaluate path costs, which may leads to the backtracking phenomenon. Although the grid-based algorithms such as A* [16] can help calculate the path cost, they frequently fall short of real-time requirement due to the complex environment characteristics.

To solve these problems, various approaches have been suggested by some academics to calculate the path length. Sun *et al.* [12] used simulated wavefronts to evaluate multiple path to get reasonable path cost. However, as the map becomes larger, more computing resources will be consumed. Li *et al.* [17] used reduced approximated generalized voronoi graphs to build a topology map of the environment to improve autonomous exploration. Nevertheless, extracting the topological information of the environment RAGVG is time-consuming and will waste a lot of memory when exploring large buildings. Dahhan *et al.* [18] proposed to use voronoi boundary for path planning. While the A* algorithm is still used to search for GVD nodes, as the dimension increases, a lot of computing resources are required. Chi *et al.* [19] proposed to use GVD feature maps to build feature trees to find the heuristic path to calculate the path cost. The process of fusing the map is time-consuming, and the real-time performance is low efficiency in actual exploration due to the characteristics of off-line operation.

In this paper, we use the information from the GVD to improve the efficiency of the autonomous exploration of the robot. A heuristic GVD frontier fusion and extraction algorithm is proposed, allowing us to swiftly obtain the fused frontiers and significantly shorten the robot motion planning decision-making time. Besides, a GVD feature nodes fusion algorithm is proposed to calculate the path cost. Overall, our method achieves great improvements in reducing robot exploration time and backtracking phenomenon.

III. METHODOLOGY

In this paper, we propose a new robot exploration strategy by quickly extracting the heuristic frontiers and reducing the backtracking phenomenon in complex environments. The system structure of GVD-based robotic autonomous exploration method is shown in Fig. 2. First, for the purpose of creating an initial map, the SLAM module is supplied to receive sensor data. The GVD nodes are then extracted from the map through the GVD construction module. Second, with the GVD information, the frontiers on the boundary are extracted according to the number of the map grid values within the radius of the GVD and fused in order to reduce the redundancy of the frontiers. Last, we acquire a GVD feature node set after extracting the GVD feature nodes. When the heuristic frontiers are received, we add the current pose of the robot and the heuristic frontiers to the GVD feature node set. Through the collision detection module, feasible edges among feature nodes are generated and a feature matrix is obtained. With the feature matrix, the robot can quickly calculate the distance between the robot and the candidate frontiers by using the Dijkstra algorithm [20]. We use the utility function to comprehensively consider the information gain and path

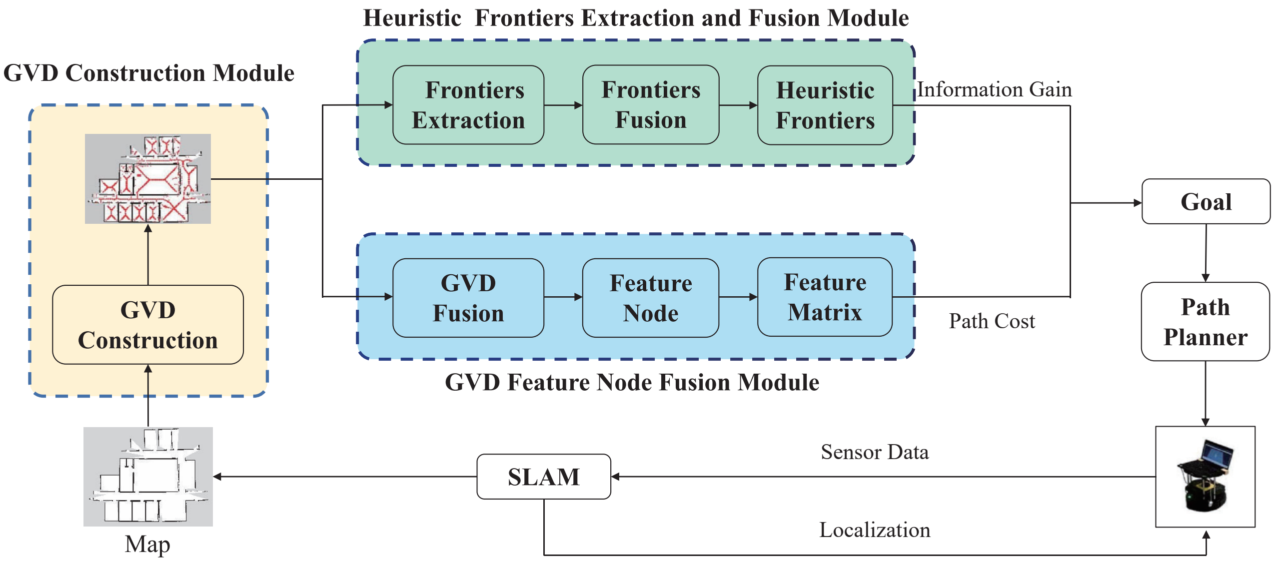

Fig. 2. Overall framework of the GVD-based robot exploration algorithm.

cost of the frontiers, and select the optimal frontier as the goal of the robotic autonomous exploration. Finally, the robot use the $move\_base$ framework for motion planning.

A. GVD Construction

When the environment information is obtained, we first extract the GVD nodes from the current map. The definition of GVD is the node with the same Euclidean distance between the nearest obstacles.

Let $\mathcal{X} \in \mathbb{R}^n$ be the state space. Let $\mathcal{X}_{obs}$ be the obstacle space and $\mathcal{X}_{free}$ = $\mathcal{X} \setminus \mathcal{X}_{obs}$ be the free space. A GVD node p is required to satisfy the following conditions:

$$\begin{cases} x_{min} = arg \underset{x_i \in \mathcal{X}_{obs}}{min} ||p - x_i|| \ , \\ \text{s.t. } num(x_{min}) \geq 2 \ , \\ p \in \mathcal{X}_{free} \ . \end{cases} \tag{1}$$

If the condition is met, the Euclidean distance between x_{min} and p is defined as the radius r. Hence, a GVD node maintain the maximal distance from obstacles in the environment. As seen in Fig. 3(a), circles with a radius equal Euclidean distance to the closest obstacles are depicted in the red area, representing the open space in the environment effectively. The blue line is composed of the GVD nodes. Here, we denote a GVD node as G_i = (x_i, y_i, r_i), and use C_i to represent a circle with (x_i, y_i) as the center and r_i as the radius. As shown in Fig. 3(a), r_i represents the Euclidean distance from the GVD node to the nearest obstacles a, b and c, and we can see that the open space in the environment is completely covered within the circle of GVD nodes. The extracted GVD nodes are shown in Fig. 3(b).

B. GVD-based Frontier Extraction

During the robot exploration, we map the GVD nodes to the current map after GVD extraction process. As depicted in Fig. 3(c), it can be found that the GVD nodes near the boundary covers the unknown area within their radius. By judging the map grid values, we can extract these GVD nodes as heuristic frontiers to guide the robot exploration. However, we can also find that there are many redundant GVD nodes around a boundary, as shown in Fig. 3(d). The redundant candidate frontiers will increase the decision-making time during the robot motion planning, making exploration inefficient. In order to reduce the redundancy of the heuristic frontiers near the boundary, we propose a GVD frontiers fusion algorithm to solve this problem. In Algorithm 1, we specify the GVD frontiers fusion algorithm. The extraction of frontiers starts from the GVD node with the largest radius G. While G is not empty, if the quantity of unexplored grids covered within the GVD node radius is larger than the threshold we set, we add G_{max_i} to the node set G_F, and delete all GVD nodes G_i located within the circle C. Otherwise, we discard the point. Up until all GVD nodes are traversed, this process iterates. Heuristic frontiers are displayed in Fig. 3(e). We found that the frontier fusion algorithm can help reduce redundant frontiers near the boundary, greatly reducing the computational complexity of the decision.

C. Frontier Selection based on GVD Feature Nodes

When the GVD frontiers extraction process is completed, the robot needs to choose the optimal frontier as the exploration goal to guide the robot exploration. In this work, we design an optimal frontier selection utility function with following formula:

$$U_f = I_f - h * N_f \tag{2}$$

Within Eq. (2), $\boldsymbol{I}_f$ represents the number of unexplored grids in the circle within the radius of the candidate frontier and $\boldsymbol{N}_f$ defines the actual path length between the current position of robot and the position of frontier. h is a weight that is a user-defined constant.

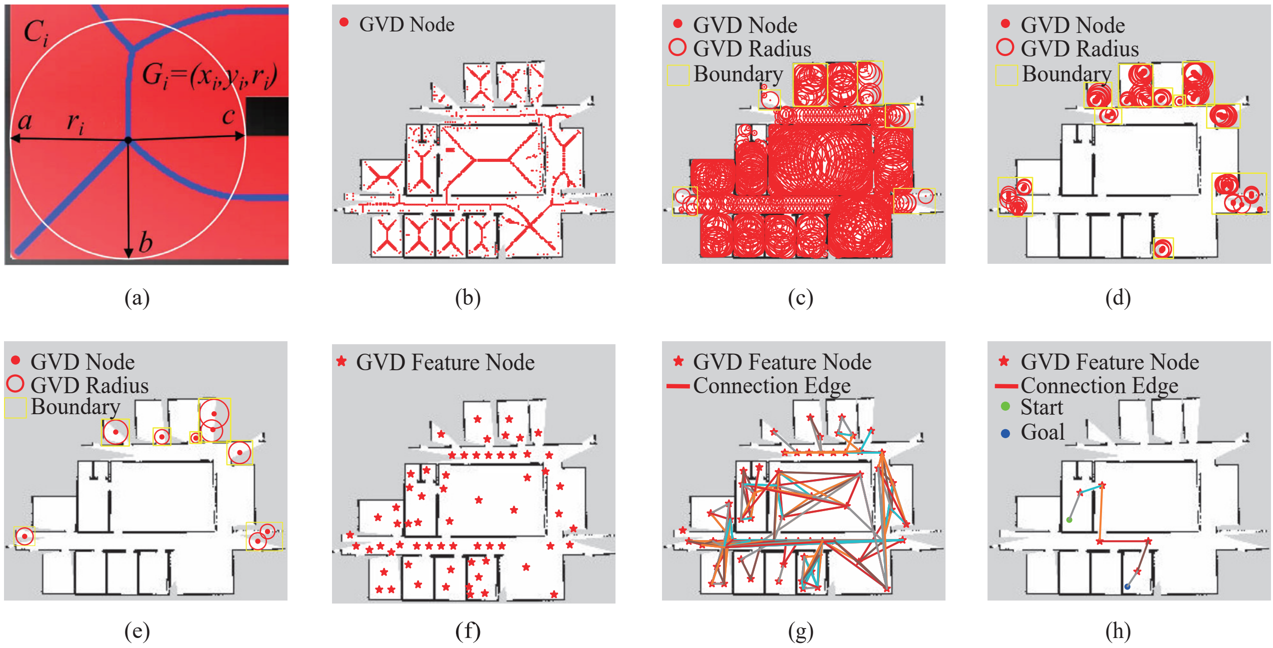

Fig. 3. Illustration of the GVD-based robot exploration algorithm. (a) shows the principle of GVD nodes construction algorithm. (b) shows the extracted GVD nodes. Red points represent GVD nodes. (c) shows GVD nodes in the map. As shown by the yellow rectangle in (c), (d) and (e), (c) shows the unknown area is covered within the radius of the GVD node of the boundary. (d) shows the redundant frontiers of the boundary. (e) shows the fused frontiers of the boundary. (f) shows the fused GVD feature nodes of the environment. Red asterisks represent GVD feature nodes. (g) shows the feasible edge connections between GVD feature nodes. Asterisks represent GVD feature nodes and lines represent the feasible edge. (h) shows the heuristic path. The green dot denote the start pose, and the blue dot denote the goal.

Algorithm 1: The Heuristic Frontiers Extraction and Fusion

Input: The map M; GVD nodes $G = (x,y,r)$; Threshold condition α; The number of unknown area values within the radius defines as Num;
Output: The Frontier set G_F

```
while G is not empty do
    max_i = arg max(r_i), ∀ r_i ∈ G
    if Num > α then
        add G_{max_i} to G_F
        for all G_i in the area of C_{max_i} do
            remove G_i from G
        end
    end
    else
        r_i = 0
    end
end
return G_F
```

As mentioned above, when evaluating the motion cost of the robot from current pose of the robot to the target frontier, two methods are usually used: one is to estimate the straight-line distance between two points, and the other is to calculate the specific path length by using the path planning method. The former ignores obstacles in the environment and the path cost is usually not accurate, which may cause non-optimal frontier selection. The latter outputs the actual path length based on the current map, however, it requires to perform path planning for each candidate frontiers, which is quite time-consuming. To address this issue, we advocate calculating the specific path length by using the GVD feature matrix. To reduce GVD redundancy, a fusion algorithm is proposed to extract the GVD feature nodes. With the GVD feature nodes, the environment features can be represented more concisely.

The GVD feature nodes fusion and feasible edge connection algorithm is displayed in Algorithm 2. The GVD feature nodes fusion starts from the GVD node G_{max} with the largest radius. While G is not empty, we add G_{max} to the feature node set G_G and delete all GVD nodes G_i located within the circle C_i. Up until all GVD nodes are covered, this process iterates. As shown in Fig. 3(f), we discovered that the GVD feature nodes fusion algorithm can assist in reducing redundant nodes within the GVD radius. After the GVD feature nodes fusion, a topology map of the environment characteristics is obtained. We add the current pose of the robot and the heuristic frontiers to the feature node set G_G, and perform collision checking on all nodes in the feature node set G_G to generate feasible connectable edges among the nodes. After that we obtain a feature matrix M_G with the distance information between each node. The feasible connection edges between nodes are shown in Fig. 3(g). Then, we utilize Dijkstra algorithm for breadth-first search to get the shortest path cost from the current position of the robot and the position the frontier and the path is shown in Fig. 3(h). After selecting

Algorithm 2: The GVD Feature Nodes Fusion and Feature Matrix Construction

Input: The map M; GVD nodes $G = (x,y,r)$; Start pose x_{start}; Goal pose x_{end}
Output: The GVD feature set G_G and its corresponding feature matrix M_G

while G *is not empty* **do**
  max_i = arg max(r_i), $\forall\ r_i \in G$
  add G_{max_i} to G_G
  for *all* G_i *in the area of* C_{max_i} **do**
    remove G_i from G
  end
end
add x_{start} and x_{end} to G_G
W is the size of G_G
Initialize M_G to a 2D zero matrix of size $W * W$
for G_{G_i}, $G_{G\_j} \in G_G$ **do**
  $free$ = $freeCheck$ (G_{G_i}, G_{G_j})
  if $free == faulse$ **then**
    $M_{G\_ij} = \infty$
  end
  else
    M_{G_ij} = $Cost$ (G_{G_i}, G_{G_j})
  end
end
return G_G and M_G

the optimal frontier, the *move_base* motion framework is utilized for robot motion planning. The environment map will be updated when the robot arrives at the frontier. Then, we extract the new exploration frontier, and this process is iterated until the map is fully constructed.

IV. EXPERIMENTAL STUDIES AND RESULTS

In this section, we conduct a series of simulation studies based on the Robot Operating System (ROS) and software to confirm the efficiency of our GVD based exploration method. For the simulation research, a laptop with Intel i7-9750H CPU and 8GB RAM, and equipped with Ubuntu 16.04 system is adopted. As a reference, the RRT-Exploration method [11] is used.

A. Frontiers extraction and selection

In this part, the experiment of candidate exploration frontiers extraction is performed based on the Robot Operating System (ROS) and the optimal exploration frontier election is conducted based on Python. As shown in Fig. 4, we conduct frontiers extraction experiment in three different environments. The viewing field and sensing distance of the laser rangefinder are the same size. The target frontiers extracted by the RRT-Exploration are shown in Fig. 4(a)–(c), while those extracted by our GVD based exploration method are shown in Fig. 4(d)–(f). Obviously, the frontiers extracted by our method can effectively reflect the boundary information, while the frontiers extracted by RRT-Exploration have a lot redundancy. Therefore, our

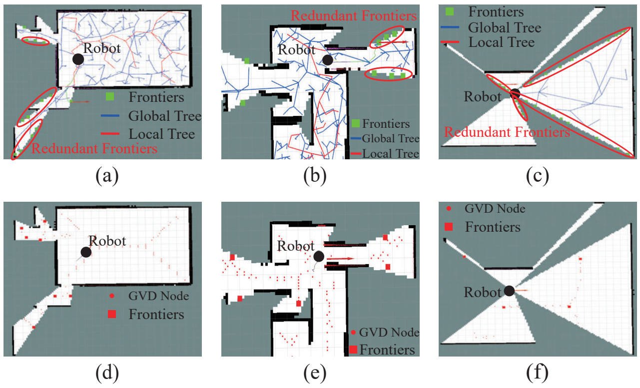

Fig. 4. Extraction of the candidate exploration frontiers. (a)–(c) represent the performance of the candidate exploration frontiers extracted by RRT-Exploration method while (d)–(f) represent the performance of the candidate exploration frontiers extracted by our GVD based exploration method. In (a)–(c), the black circle is current pose of the robot in the environment, and the green squares represent the candidate frontiers extracted by RRT-Exploration. The local tree is represented by the red lines, the global tree is represented by the blue lines. In (d)–(f), the black circle is current pose of the robot in the environment, and the red dots represent the GVD nodes. The red squares denote the candidate frontiers extracted by our method.

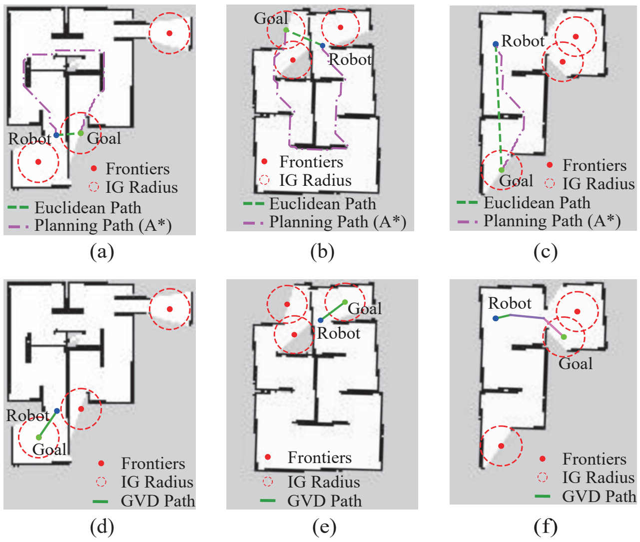

Fig. 5. Illustration of the optimal exploration frontier selection. (a)–(c) state the results of the optimal exploration frontier selected by RRT-Exploration method while (d)–(f) state the results of the optimal exploration frontier selected by our GVD based exploration method. The blue dot represents current pose of the robot in the environment. The red dots represent the frontiers extracted by our GVD based exploration method and the red circles represent the information gain radius(IG Radius). The green dot denote the optimal exploration frontier as the goal. Dotted lines represent the Euclidean path. Solid lines represent the GVD path. Dash-dotted lines represent the A* path.

method can effectively reduce the computational complexity in robot decision-making process.

The optimal frontier selection experiment is conducted in three different environments, as shown in Fig. 5. The current robot pose and the known map are given during the selection process. The heuristic frontiers are extracted by our algorithm. The optimal exploration frontier selected by the RRT-Exploration is shown in Fig. 5(a)–(c), while that selected by our GVD based exploration method is shown in Fig. 5(d)–(f). As shown in Fig. 5(a)–(c), the actual path length of the distance from the robot to the goal planned by the A* [16] algorithm is much larger than the Euclidean path. Obviously, the exploration frontier selected by RRT-Exploration ignore the environmental characteristics, so that the robot cannot select the optimal exploration frontier, which leads to the backtracking phenomenon. While, our method incorporates the actual path cost so that can efficiently evaluate the information gain of each exploration frontier. In conclusion, our method can effectively select the optimal frontier, so as to reduce the backtracking phenomenon during the robot exploration process.

B. Exploration performance

In this part, we conduct a series of exploration performance tests in the Robot Operating System (ROS). In the simulation environment, we randomly set up three typical scenarios and perform the experiments in these scenarios, as shown in Fig. 6.

In scenarios 1 and 2, their corresponding size are $10m * 10m$, and the viewing field and sensing distance of the laser rangefinder are both adjusted to $180°$ and $8m$, respectively. In scenario 3, the corresponding size is $20m * 12m$. The laser rangefinder sensing distance is set at $10m$ and the laser rangefinder field of view is $270°$. For each experimental scenario, we conduct five sets of repeated experiments. In Fig. 6, we recorded the trajectories of the robotic exploration process in three different environments. In Fig. 7, we record the time and path length of the autonomous exploration by the robot.

In the first map, time of the exploration reduce 15.0% and path length of the exploration reduce 12.9% on average, respectively, by comparing our method with the RRT-Exploration method. During the experiments of RRT-Exploration, the robot misestimated the path length between the robot and the candidate frontiers three times, which results in the backtracking problem and decrease the exploration efficiency.

In the second map, our method reduces the path length by 17.0% and the time by 18.6% on average. We can find that the trajectory in the middle of the map has a lot of redundancy. Limited to the disadvantage of random sampling of the RRT tree, the extraction of candidate frontiers by the RRT-Exploration is slow. The robot goes to the next environment before one environment is fully explored, and the path cost is not reasonably used to select a optimal goal during the exploration process, resulting in a large backtracking phenomenon.

In the third map, we ensure that the robot exploration does not generate backtracking phenomenon, the path length explored by our method is reduced by 4.5%, but the exploration time is much lower than that of RRT-Exploration, which is reduced by 22.6%. The experimental results confirm that our method can also quickly extract frontiers in a large map environment. However, during the robot exploration of RRT-Exploration, the global tree grows

Fig. 6. Trajectories in three typical scenarios during the robot exploration. The blue dot denote the start pose. The left column is the exploration trajectories using the RRT-Exploration method, and the right column is the exploration trajectories using our GVD based exploration method.

slowly, and only local trees can be used to obtain frontiers, which results in low efficiency of the frontiers extraction.

In summary, compared with the RRT-Exploration, our method has a significant improvement in both exploration time and path length. Our method can effectively solve the disadvantage that RRT-Exploration cannot obtain exploration frontiers in time because of the randomness of sampling. Furthermore, our method can effectively reduce the redundancy of exploration frontiers. During the robot exploration, the robot can use the specific path cost information in time to ensure the optimal decision-making, so as to reduce the backtracking phenomenon. Overall, the two metrics of the exploration path length and time provide conclusive evidence of the efficiency of our method.

V. CONCLUSION

In this paper, a Generalized Voronoi Diagram based autonomous exploration method for mobile robots is proposed. First, we extract the GVD nodes of the environment in real time. Second, a GVD frontiers fusion algorithm is proposed to quickly generate the fused heuristic frontiers. Then, we propose a GVD feature nodes fusion algorithm and use the collision detection to get the GVD feature matrix. By searching the GVD feature matrix, we can

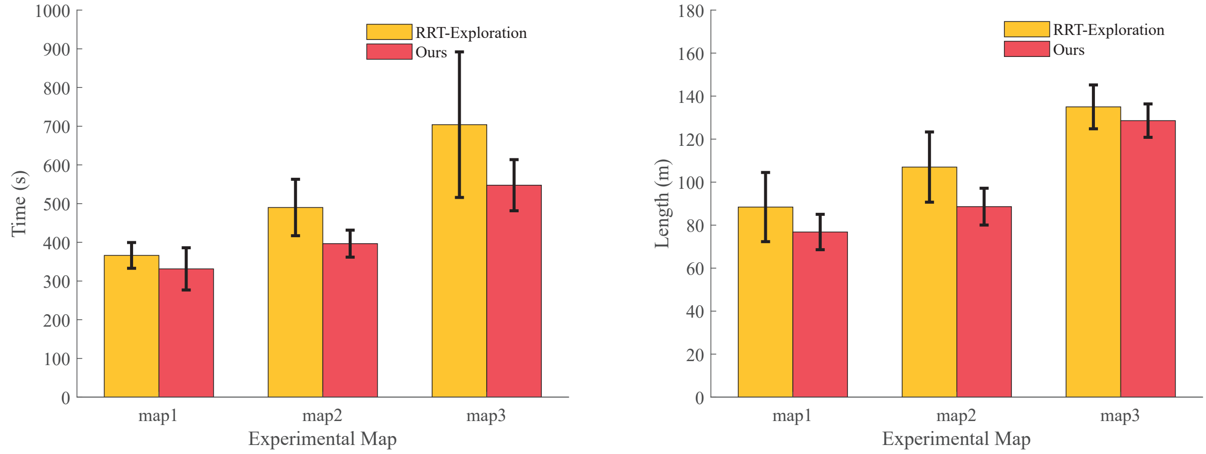

Fig. 7. Comparison of the robot exploration performance. The left figure shows the robot exploration time and the right figure shows the robot exploration path length on average.

calculate the specific path length and then select the optimal frontier as the exploration goal. Finally, the robot explore the target area by utilizing the $move\_base$ motion framework until the environment map is fully constructed. We conducted experimental studies in different scenarios. The experimental results indicate that our method can effectively reduce the exploration time and the backtracking phenomenon in the process of robot exploration, improving the efficiency of the robot exploration, and demonstrate the effectiveness of the method.

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